

# **Cloud Computing**

# Basics of Computers

- **Mainframe:**

- A mainframe is a large, expensive, powerful server that can handle hundreds or thousands of connected users/servers simultaneously.
- For example a single mainframe server of IBM's Z series can provide the equivalent computing throughput of at least 500 servers.





# Basics of Computers

3

- **Mainframe leasing model:**
  - The customers were charged on monthly basis for the use of hardware such as CPU, memory and peripheral devices.
  - The software (compilers, editors etc.) usage was charged for the time of usage.

# Basics of Computers

- **Server:**
  - A server is a computer which provides services to other computers and/or devices connected to it.
  - Services provided by a server include the controlled access to hardware and software resources and storage.
  - A server can support hundreds and thousands of simultaneous users.



# Basics of Computers

5

- **Servers are available in a variety of sizes and types:**
  - Web server: stores websites and web apps and provides them on your desktops and mobiles through web browsers.
  - Domain Name Server (DNS): Stores domain names and the corresponding IP addresses.
  - Database server: Hosts database and provides access to data and provides data manipulation functionality.



# Basics of Computers

6

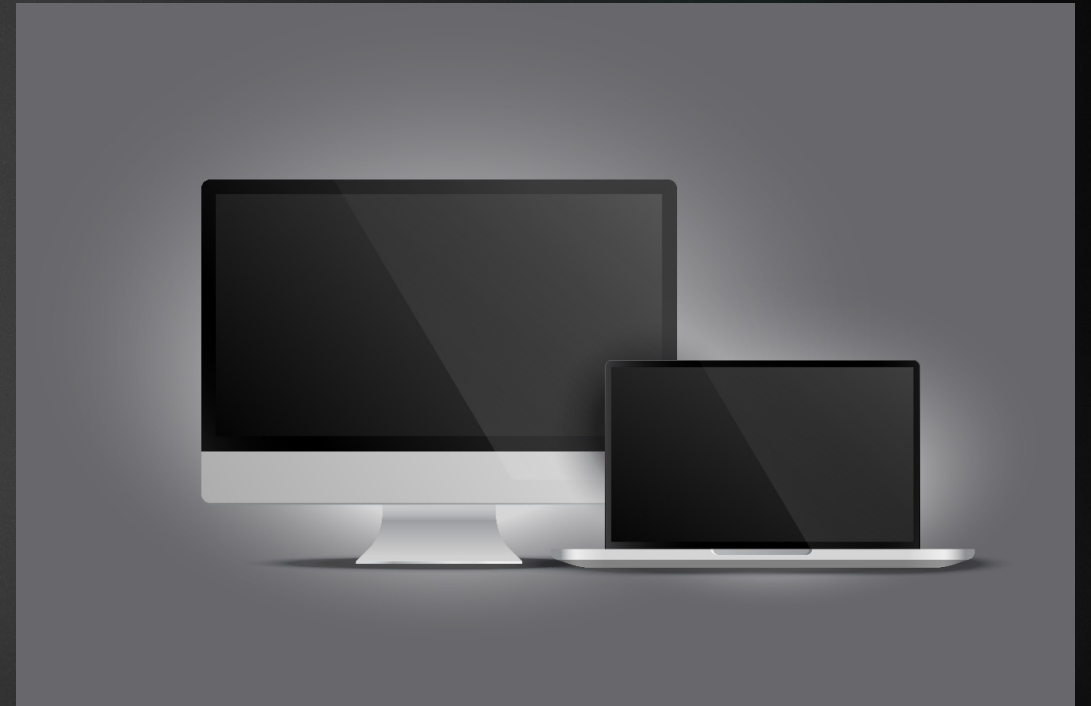




# Basics of Computers

7

- **Desktop**
- A desktop is a computer which is designed to remain in a stationary position. It is used as a personal computer.
- Performs all of its activities such as
  - Input
  - Processing
  - Output
  - Storage by itself.
- Intended to be used by one person at a time.



# Basics of Data Communication

- **Data Communication:**

- Exchange of data over some transmission medium between two devices.
- **The following factors are essential for data communication:**
  - Data must be delivered to correct destination.
  - Data must not be altered during transmission.
  - There must be timely delivery of the data.



# Basics of Data Communication

9

- ▶ There must not be uneven delay among the packet arrival time during audio or video transmission.
- **Components:**
  - Message: The data to be sent. Can be text, numbers, pictures, audio and video.
  - Sender
  - Receiver

# Basics of Data Communication

10

- **Transmission medium:** The physical path through which a message travels from sender to receiver.
- **Protocol:** The set of agreed-upon communication-rules between sender and receiver devices.
- Two devices can be connected but not communicating without a protocol.



- **Data Representation:**

- Text: Represented by bit pattern called code e.g.; Unicode and American Standard Code for Information Interchange (ASCII).
- Numbers: Directly converted binary of the number. ASCII is not used to represent numbers.

# Basics of Data Communication

12

- **Images:** Sent as binary patterns. Image is represented by a matrix of pixels. Pixel is a small dot. Each pixel is assigned a bit pattern on the basis of color.
- **Audio:** A continuous stream of data. Different from text, numbers and images.
- **Video:** Can be a continuous stream or a sequence of image combinations.



- **Simplex:**
  - Unidirectional communication in which either one of the sender or receiver device can transmit. For example: key board, monitor etc.
- **Full Duplex:**
  - Both devices can send and receive at the same time. The transmission medium should provide separate paths (channels) for the transmission of each device. For example telephone conversation is full duplex.

# Basics of Computer Networking

14

- **Computer networking was conceived in 1960s soon after the invention of computers.**
- **A network is a collection of computers and devices connected together through transmission media.**



# Basics of Computer Networking

15

- **Devices:**
  - **Hosts:** Large computers, desktops, laptops, cellular phone or security system.
- **Connecting devices:**
  - **Router:** A device which connects the network with other networks.
  - **Switch:** A device which connects devices within the network.
  - **Modem:** A device which changes the form of data (modulates-demodulates).

# Basics of Computer Networking

16

- **Network Criteria:**
  - Performance: It is often evaluated by two metrics:
  - Throughput (bulk of data transmitted in unit of time) and delay.
  - Increasing the throughput may increase the congestion and hence increase the network delay.



# Basics of Computer Networking

17

- ▶ The transit time (message travel time) and response time (time between inquiry and response) indicate the network performance also.
- ▶ Reliability: It is measured in terms of frequency of network failure, time to recover from a failure and robustness from disasters.
- ▶ Security: Protecting data from unauthorized access and damage, and implementation of security policies and procedures for recovery from breaches and data losses.

# Basics of Computer Networking

18

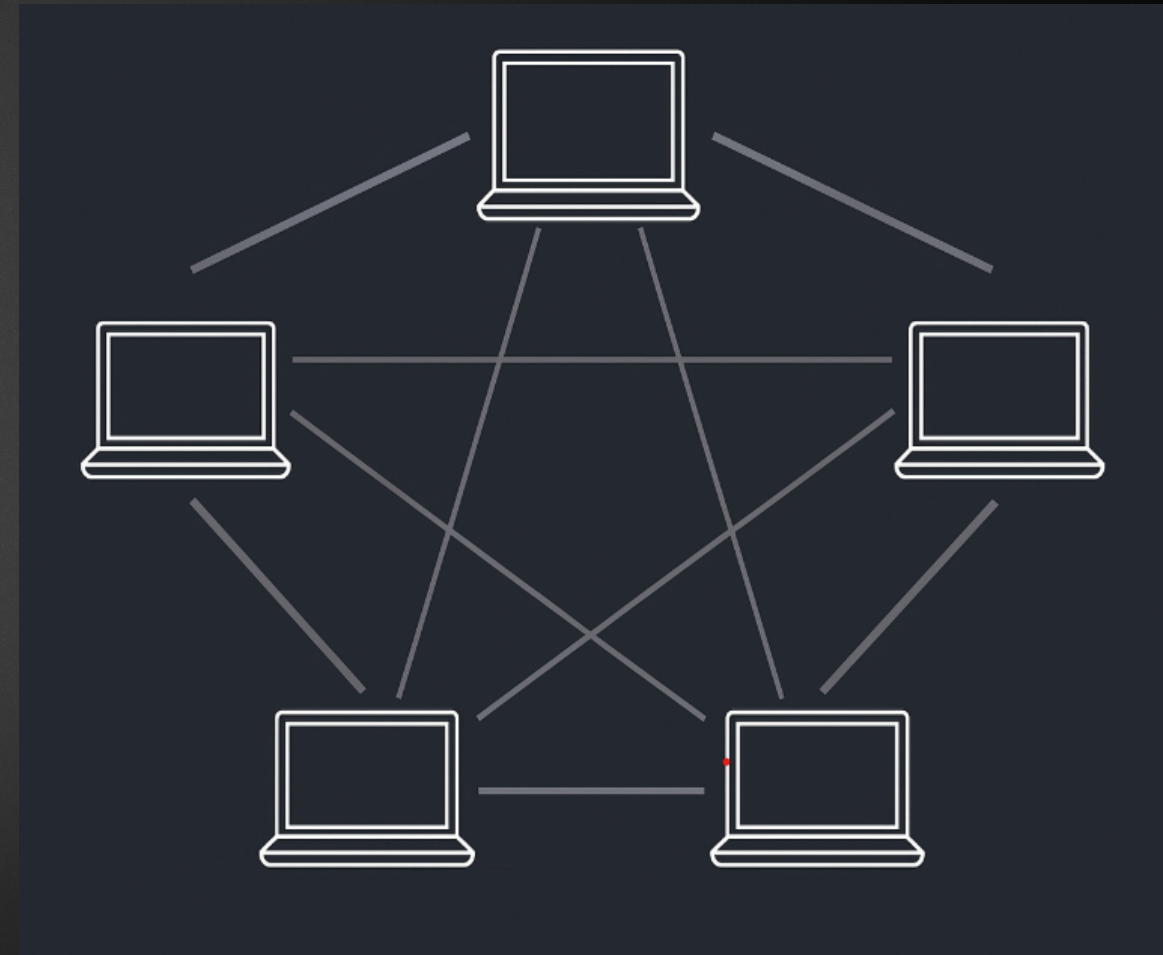
- **Physical Structures:**

- Network Connections: Communication can only take place if the devices are simultaneously connected to the same communication-path or link or connection.
- A link can be dedicated link (Point to Point) or shared among devices (multipoint).



# Basics of Computer Networking

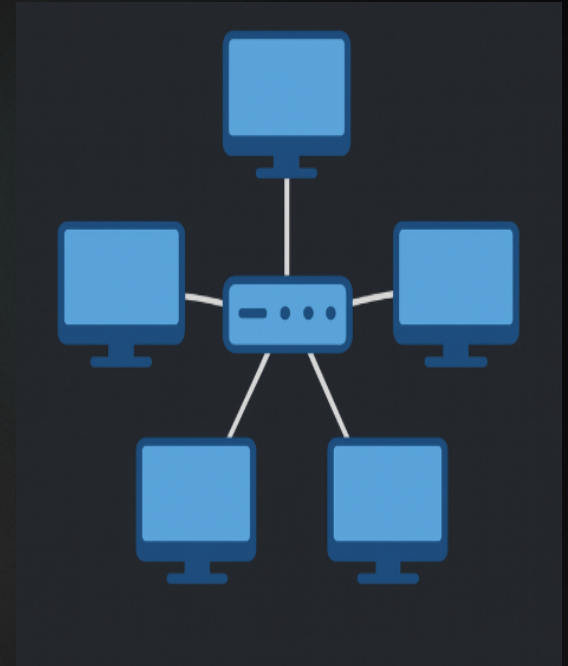
- **Physical Topology:**
  - **Mesh:** Every device has a dedicated point to point link to every other device.
  - Advantage: robustness of network from failure of any link.
  - Disadvantage: The bulk of cabling involved.



# Basics of Computer Networking

20

- **Star:** All devices are connected to a central device. Unlike mesh, there is no direct traffic between any two devices but through the central device such as hub.
- Advantage: Requires only one I/O port in each device as compared to mesh.
- Disadvantage: If the central device fails, the whole network fails.





# Basics of Computer Networking

21

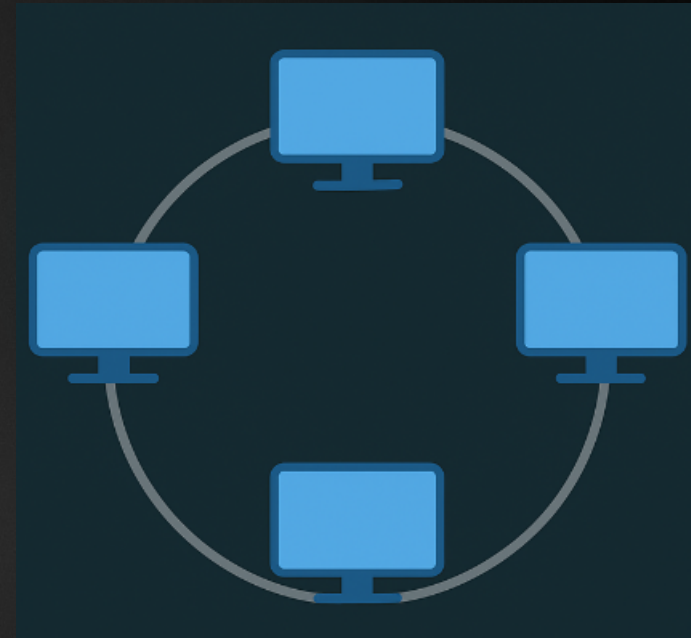
- **Bus:** A multipoint topology in which one long cable is used as a network backbone.
- Advantage: Ease of installation. Requires less cabling than mesh and star.
- Disadvantage: Difficult to extend, signal drops along the length of cable results in limited number of connections, breaking of backbone cable isolates the network segments and introduces noise.



# Basics of Computer Networking

22

- **Ring:** The devices are connected in the form of ring. Each device acts as repeater.
- **Advantages:** Easy to expand and alter the network. **Disadvantage:** Failure of a single device can disable the entire network, transmitting device needs to retain the token signal to perform transmission which slows down the data rate.





# Network Types

23

- **Local Area Network (LAN):** It is a privately owned network and has a scope of an office, building or a campus. A LAN can even extend throughout a company.
- Each host in a LAN has a unique identifier or address.
- The communication packets between any two hosts in a LAN contain the source and destination hosts' addresses.
- **Local Area Network (LAN) key features:**
  - Media type: wired/wireless, twisted pair/cable/fiber, radio, infrared
  - Topologies: Bus, Star, Mesh, Ring, Tree
  - Bit rate: from 1Mbps to 1Gbps

# Network Types

24

- **Typical LANs:**
  - Ethernet (CSMA/CD): Carrier Sense with Multiple Access with Collision Detection (retransmission after collision detection)
  - Local Talk (CSMA/CA): CSMA with Collision Avoidance (reserve the media before transmission)
  - Wireless LAN: IEEE 802.11, Range: < 100 m, Speed: 2Mbps



# Network Types

25

- **Typical LANs:**
  - Token Ring: A token travels around the ring, it must be retained by the sender computer to send a single packet, 4,6 or 100 Mbps
  - FDDI: Token ring with fiber optic cable, 100 Mbps
  - ATM: Star based, uses switch, multiple devices can communicate simultaneously, 25, 45, 155, 600+ Mbps
- Unicast, Broadcast, Multicast

# Network Types

26

- **Wide Area Network (WAN):** A network that spans large geographical area such as town, cities, states or even countries. Usually interconnects multiple LANs.
- Unlike LAN which is owned by the user organization, a WAN is normally created and run by communication companies. It is leased to the user organizations.
- **Wide Area Network (WAN) Types**
  - P2P WAN: Connecting two devices through wired or wireless media e.g.; connecting two LANs to form a private internet or internetwork of a company.
  - Switched WAN: A network with more than two ends. It is a combination of several P2P WANs connected by switches.



# Network Types

27

- **Metropolitan Area Network (MAN):** It is a computer network covering a large geographical area bigger than LAN and smaller than WAN.
- Diameter: 5 to 50 km, several buildings or a whole city.
- MAN is not owned by a single organization generally just like WAN. The MAN equipment are usually owned by a service provider.
- MAN usually provides high speed connectivity to allow sharing regional resources.

# Wired LAN vs. Wireless LAN

28

## ▶ Wired LAN

- ▶ Medium: Wires
- ▶ Broadcasting and multicasting possible when required
- ▶ Physical connection to network
- ▶ Hosts are connected through link layer switch
- ▶ Connection to other networks through router

## ▶ Wireless LAN

- ▶ Medium: Air
- ▶ All devices are broadcasting
- ▶ No physical connection to network
- ▶ No link layer switch exists
- ▶ Connected to other networks through access point (a device that connects a wireless and wired network)



- It is a wireless LAN standard by IEEE that covers physical and data-link layers
- Synonyms: WiFi, Wireless LAN
- Basic architecture consists of an access point (AP) and capable devices connected to AP
- In the absence of AP, the wireless devices connect to form adhoc network
- Multiple overlapping APs are used to cover a larger area
- ▶ A device is connected to only one of the nearest APs CSMA/CA is used. The sender sends a Request To Send (RTS) packet, the receiver sends Clear To Send (CTS) packet, the sender sends data after receiving CTS, the receiver sends acknowledgement, the other senders can send now.
- ▶ If no CTS is received, the sender marks it as a collision

# IEEE 802.11

30

- ▶ 802.11 a, b, g, n
- ▶ 802.11a: 50 feet, 22 Mbps
- ▶ 802.11b: 100 feet, 11 Mbps
- ▶ 802.11g: 100 feet, 54 Mbps
- ▶ 802.11n: 50 feet, 700 Mbps
- ▶ 802.11ac 100 feet, 1.3 Gbps
- ▶ 802.11ax medium, 9.6 Gbps
- ▶ 802.11be medium, 46 Gbps



- ▶ It is a wireless LAN technology that provides short distance connectivity to devices which have different functionalities for example, mobile phones, headsets, notebooks, desktops, computer peripheral devices, cameras and even the home appliances.
- Multiple devices can be connected through Bluetooth to form a piconet.
- Bluetooth supports:
  - Voice and data transmission
  - Adhoc networking for up to 10 meters
- Multiple devices can be connected through Bluetooth to form a piconet
  - IEEE standard 802.15 covers the Personal Area Network (PAN) using Bluetooth for an area covering a room

- **Versions:**

- 1.x: up to 1Mbps, obsolete
- 2.x: up to 3 Mbps, improved pairing capability between devices from different manufacturers

- **Versions:**

- 3.x: up to 24 Mbps using WiFi 802.11
- 4.x: Up to 1 Mbps, works seamlessly with 4G, works with data collection from sensors an internet of things (IoT)
- 5.x: Up to 2Mbps, stronger connection, better power efficiency support multiple devices



- WiMax stands for Worldwide Interoperability for Microwave Access
- Provides wireless access to Internet for:
  - Homes and offices when the wired access is either not available or is expensive (fixed WiMAX)
  - Mobile phones (mobile WiMAX)
- ▶ Fixed WiMAX requires the installation of antennas at the premises of the subscriber to receive and send the data from the base station of Internet provider.
- ▶ Mobile WiMAX users move from one place to another while connected to the base station of Internet provider .

- WiMAX is the result of IEEE 802.16 project. It is a standard for wireless WAN (or MAN). The subscriber station may be tens of kilometers away from the base station of the provider.
- Remember that 802.11 is the standard for wireless LAN.
- Uses 48-bit MAC address of subscriber station and base station at Data-link layer
- Connection oriented protocol. Each connection has a unique id and hence there is no address field in the frame of WiMAX
- Full duplex communication



# Evolution of Cellular Networks

35

- Cellular network or telephony is a radio-based technology
- Radio waves are electromagnetic waves propagated by antennas
- Note: Antenna is a transducer device which converts the altering current into radio waves and vice versa
- 7 billion mobile connections
- 25 billion interconnected devices count predicted in 2020
- Over 100 billion downloads completed in 2013, 270 billion expected in 2017, 370 billion downloads across the world
- The base stations receive from and transmit to cellular phones.
- Cellular Networks have evolved from first generation (1G) to fifth generation (5G)
- Let us briefly look at these generations...

# Evolution of Cellular Networks

36

- 1G

- Invented around 1980.
- First implementation in Tokyo (Japan)
- Based upon analog technology
- Expanded to cover all the population of Japan in few years

- 1G

- Not secure
- Anyone with an all-band radio can listen to calls and get the phone number of the subscriber
- Analog mobiles were larger in size and heavy in weight



# Evolution of Cellular Networks

- 2G

- Invented in 1991, implemented first time in Finland
- Technologies: Global System for Mobile (GSM) Communication, General Packet Radio Service (GPRS), Code Division Multiple Access (CDMA) [digital signal] and Enhanced Data Rates for GSM Evolution (EDGE)

- 2G

- Short Messaging Service (SMS), Multi-Media Messaging Service (MMS)
  - Typical data rate: 100 Kbps
  - Email, Web browsing, Camera phones

# Evolution of Cellular Networks

- 2G

- Signal strength decay problem, performance degrades with the rise in number of users in a cell (area maintained by a base station)

- 3G

- From 2000 to 2010
- Technologies: CDMA, WLAN, Bluetooth, Universal Mobile telecommunication Systems (UMTS), High Speed Downlink Packet Access (HSDPA)



# Evolution of Cellular Networks

39

- 3G

- Features: Global Roaming Clarity in voice calls, Fast Communication, Internet, Mobile T.V, Video Conferencing, Video Calls, Multi Media Messaging Service (MMS), 3D gaming and Multiplayer-Gaming, smart phones

- 3G

- Typical data rate: Up to a few Mbps
- Expensive mobile phones, battery life issue

# Evolution of Cellular Networks

- 4G
  - Since 2010
  - Technologies: Long Term Evolution (LTE) Standard based on the GSM/EDGE and UMTS/HSPA, Multiple In Multiple Output (MIMO) smart antenna technology, Orthogonal Frequency Digital Multiplexing (OFDM), WiMAX
  - Typical data rate: Up to a few tens of Mbps
  - MAGIC: Mobile multimedia–Anytime anywhere–Global mobile support–Integrated wireless solutions–Customized personal service
  - Maintaining data rate is an issue, not fully implemented in all the world, battery consumption is a bigger problem than 3G



# Evolution of Cellular Networks

41

- 5G
  - Since 2012
  - Technologies: MIMO, Network Slicing (creates virtual networks for different use), OFDM (Orthogonal Frequency Division Multiplexing), Small Cells.
  - Faster data rate than 4G ( $> 1\text{Gbps}$ ), higher data rate at cell edges
  - Low latency, Faster Speed, Supports more devices, better coverage